Efficacy for Differentiating Nonglaucomatous Versus Glaucomatous Optic Neuropathy Using Deep Learning Systems

HEE KYUNG YANG, YOUNG JAE KIM, JAE YUN SUNG, DONG HYUN KIM, KWANG GI KIM, AND JEONG-MIN HWANG

• PURPOSE: We sought to assess the performance of deep learning approaches for differentiating nonglaucomatous optic neuropathy with disc pallor (NGON) vs glaucomatous optic neuropathy (GON) on color fundus photographs by the use of image recognition.

• DESIGN: Development of an Artificial Intelligence Classification algorithm.

• METHODS: This single-institution analysis included 3815 fundus images from the picture archiving and communication system of Seoul National University Bundang Hospital consisting of 2883 normal optic disc images, 446 NGON images, and 486 GON images. The presence of NGON and GON was interpreted by 2 expert neuro-ophthalmologists and had corroborated evidence on visual field testing and optical coherence tomography. Images were preprocessed in size and color enhancement before input. We applied the convolutional neural network (CNN) of ResNet-50 architecture. The area under the precision-recall curve (average precision) was evaluated for the efficacy of deep learning algorithms to assess the performance of classifying NGON and GON.

• RESULTS: The diagnostic accuracy of the ResNet-50 model to detect GON among NGON images showed a sensitivity of 93.4% and specificity of 81.8%. The area under the precision-recall curve for differentiating NGON vs GON showed an average precision value of 0.874. False positive cases were found with extensive areas of peripapillary atrophy and tilted optic discs.

• CONCLUSION: Artificial intelligence–based deep learning algorithms for detecting optic disc diseases showed excellent performance in differentiating NGON and GON on color fundus photographs, necessitating further research for clinical application. (Am J Ophthalmol 2020;216:140–146. © 2020 Elsevier Inc. All rights reserved.)

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GLAUCOMA IS THE LEADING CAUSE OF BLINDNESS worldwide that slowly advances from an early to late stage.[1] Early diagnosis and treatment of glaucoma is important to reduce the risk of progressive and irreversible visual loss.[2] Meanwhile, nonglaucomatous optic neuropathy with optic disc pallor (NGON) is caused by various intracranial, systemic, and hereditary causes that require prompt neuroimaging and evaluation for detection of an underlying treatable cause.[3,4] Moderate to severe glaucoma may also end up with secondary optic disc pallor in advanced stages.[5–7] Therefore it may be hard to distinguish between different etiologies solely based on morphologic assessment of the optic disc.[3,8] Moreover, certain conditions, such as physiologic temporal pallor, may frequently be misinterpreted as optic disc pallor in normal subjects.[9] The manual interpretation of fundus photographs is also subject to interobserver variability and varying image qualities with different settings.[10,11]

Current trends of image analysis in medical imaging are mainly focused on deep learning.[12] Deep learning methods have been validated in the detection and grading of diabetic retinopathy, age-related macular degeneration, and glaucoma, and have shown good efficacy compared with the manual interpretation of standard funduscopic images.[13–17] Meanwhile, the morphologic assessment of NGON is more complex because of the lack of large public data sets and low reliability of manual detection compared with diabetic retinopathy and glaucoma.[3,13,14,16]

In this study, we aimed to develop a machine learning–based algorithm for the detection of glaucomatous optic neuropathy (GON) and NGON in patients with various etiologies. We demonstrated the performance of the deep learning system to differentiate GON from NGON and normal optic discs using preprocessing and convolutional neural network (CNN)–based architectures.[17–20]

METHODS

• DATA ACQUISITION: We collected 3815 fundus images from the picture archiving and communication system of Seoul National University Bundang Hospital. Subjects with indistinguishable retinal images related to media opacity, such as corneal opacity, cataract, and abnormal

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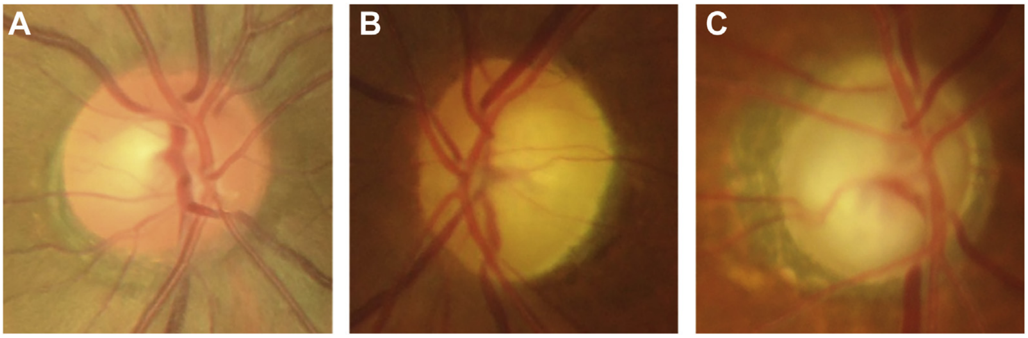

FIGURE 1. Representative figures of the cropped images with a 10% margin around the optic disc. In the cropped images, the pixel values were resampled to an image resolution of 256 × 256. (A) Normal optic disc. (B) Nonglaucomatous optic neuropathy with optic disc pallor. (C) Glaucomatous optic neuropathy.

retinal findings that obscure the detection of optic disc morphology, such as macular degeneration or retinal hemorrhage, were excluded.[11] The images consisted of 2883 normal optic disc images, 446 NGON images, and 486 GON images. Normal optic disc images were collected from patients who visited the Health Promotion Center of Seoul National University Bundang Hospital from 2013-2014. All images of NGON and GON were collected from the electronic medical record database of patients with the diagnostic codes of optic neuropathy and glaucoma from 2003-2017. The presence of NGON and GON was interpreted by 2 expert neuro-ophthalmologists and confirmed by visual field testing and optical coherence tomography (OCT) using either Stratus OCT (Carl Zeiss Meditec, Jena, Germany) or Spectralis OCT (Heidelberg Engineering, Heidelberg, Germany).

The inclusion criteria for GON were as follows: (1) a clinical diagnosis of glaucoma with progressive optic disc change, such as a vertical cup-to-disc ratio ≥0.7, asymmetry of cup-to-disc ratio ≥0.2, or the presence of focal thinning or notching; (2) global or localized thinning of the circumpapillary retinal nerve fiber layer (RNFL) with abrupt color scale change on OCT; and (3) an associated glaucomatous visual field defect measured with a Humphrey field analyzer showing a pattern standard deviation with $P < .05$ which is repeatable on ≥2 consecutive examinations. The inclusion criteria for NGON were: (1) optic nerve pallor caused by compression of the anterior visual pathway, demyelination, inflammation, ischemic optic neuropathy, toxic optic neuropathy, traumatic optic neuropathy, and hereditary optic neuropathy that were confirmed by brain magnetic resonance imaging and extensive laboratory investigation including genetic tests; (2) a pattern standard deviation with $P < .05$ measured with a Humphrey visual field analyzer; (3) intraocular pressure ≤20 mm Hg; and (4) thinning of mean circumpapillary RNFL thickness or temporal circumpapillary RNFL thickness below normal limits with $P < .01$. Inclusion criteria for healthy normal eyes included intraocular pressure ≤20 mm Hg with no history of increased intraocular pressure and the absence of glaucomatous or nonglaucomatous optic disc appearance on fundus photography.

Diagnosis of the 446 fundus photographs with NGON included acquired causes of optic neuropathy by compression and/or other brain lesions involving the visual pathway (n = 147), Leber hereditary optic neuropathy (n = 92), demyelination/inflammation (n = 50), chronic stage of nonarteritic ischemic optic neuropathy (n = 29), autosomal dominant hereditary optic atrophy (n = 24), toxic optic neuropathy (n = 22), traumatic optic neuropathy (n = 9), and unknown causes of optic atrophy (n = 73). The research was approved by the Institutional Review Board of Seoul National University Bundang Hospital and adhered to the tenets of the Declaration of Helsinki.

• **DATA AUGMENTATION:** We used 900 images for the training set, 240 images for the validation set, and 2675 images for the test set. The images of the training, validation, and test sets were partitioned randomly and images from the same patient were not included in >1 set. The training set consisted of 300 normal optic disc images, 300 NGON images, and 300 GON images. The validation set consisted of 80 normal optic disc images, 80 NGON images, and 80 GON images. The test set consisted of 2503 normal images, 66 NGON images, and 106 GON images.

The data in the training set were insufficient to train the deep learning model, and the ratio between normal optic discs, NGON, and GON were not equal in the data set. Therefore, we supplemented the training data by augmenting the number of images.[21,22] The training set was augmented by 20-fold via a random combination of flip, rotation, translation, and scale adjustment. Augmentation was not applied to either the validation set or the test set. Based on image preprocessing and data augmentation, we finally included 18 000 images with visible optic discs for training data, consisting of 6000 normal optic disc images, 6000 NGON images, and 6000 GON images.

• **PREPROCESSING:** The optic disc was the region of interest within the whole color fundus photograph. Therefore,

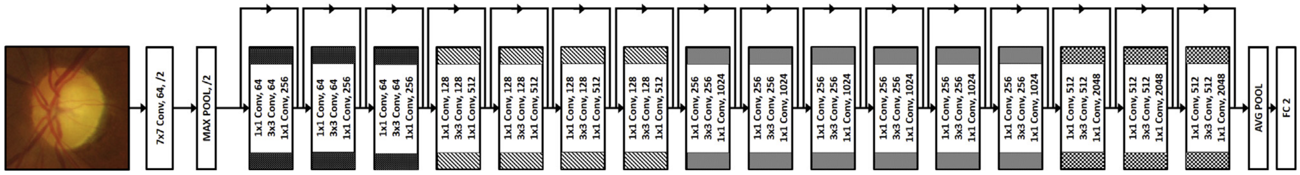

FIGURE 2. Convolutional neural network based ResNet-50 architectures of the artificial deep learning system.

TABLE. Results of the Convolutional Neural Network–Based Deep Learning Models Used for Inference on the Test Data to Detect Normal Optic Discs, Nonglaucomatous Optic Neuropathy with Optic Disc Pallor, and Glaucomatous Optic Neuropathy

	CNN (Normal)	CNN (NGON)	CNN (GON)	Accuracy (%)	Overall Accuracy (%)
Normal	2495	3	5	99.7	99.1
NGON	0	57	9	86.4	
GON	1	7	98	92.5	

CNN = convolutional neural network; GON = glaucomatous optic neuropathy; NGON = nonglaucomatous optic neuropathy with optic disc pallor. Values are the number of fundus photographs in the test data of each group.

we automatically detected the optic disc region and cropped it to exclude unnecessary areas other than the optic disc using the threshold algorithm, because the optic disc has a stronger contrast compared with surrounding tissues. We separated the image into color channels and used the red channel image to binarize each pixel with a threshold intensity of 200. A cluster of pixels with a threshold intensity >200 were labeled together, and the size of each label was compared. The label with the largest area was assumed to be the optic disc. We cropped the images with a 10% margin around the optic disc label of the original color image. In the cropped images, the pixel values were scaled to a range of 0-1, and then resampled to a resolution of 256 × 256. During resampling, interpolation was performed using the bicubic method (Figure 1).[23,24] All preprocessing steps were performed automatically.

• **DEEP LEARNING ARCHITECTURE:** Deep learning was implemented using the Keras Application Programming Interface in TensorFlow (tensorflow.org) running under the Ubuntu operating system. We used the ResNet-50 convolutional neural network to classify the 3 groups: normal, NGON, and glaucoma. The functionality of ResNet is "skip connection" that bypasses several convolutional layers at a time. Each bypass generates a residual block and the convolution layer predicts the residual added to the input tensor of the block, which promotes learning.[13,14]

Figure 2 shows the detailed structure of the ResNet-50 architecture model. The model was learned under the condition with a minibatch gradient descent of 40 size, an Adam optimizer learning rate of 0.0001, a dropout rate of 0.5, and an epoch of 300.[18,25,26] The model used the initial weights as pretrained weights from the ImageNet dataset of >15 million labeled images for transfer learning. Transfer learning is a technique that uses the weight values of other trained data as an initial value and has proven to be a highly effective technique when the data are limited.[17,18]

We verified the performance of the model by applying the trained ResNet-50 model to the separately constructed test data.

• **CLASS ACTIVATION MAP:** We generated a heatmap to visualize the prediction process of the trained model for interpretation of predictions using the class activation map (CAM). CAM can be used to visualize where the trained model influences the prediction process. To obtain the CAM, we extracted the feature maps by the final convolutional layer. We then obtained the CAM used for classifying the images by taking the weighted sum of the feature maps of the final classification layer.[19]

RESULTS

AS A RESULT, THE CONFUSION MATRIX FOR THE TEST DATA is given in the Table that presents the accuracy rates of classification for the 3 groups of normal optic discs, NGON, and GON, respectively. The accuracy of detecting normal optic discs, NGON, and GON were 99.7%, 86.4%, and 92.5%, respectively, and the overall accuracy was 99.1%.

We used precision-recall (P-R) curves to measure the P-R tradeoff of our algorithm because of the imbalanced data sets and reported sensitivity and specificity on the receiver operating characteristic curve. The P-R curves for identifying NGON and GON are plotted in Figure 3.

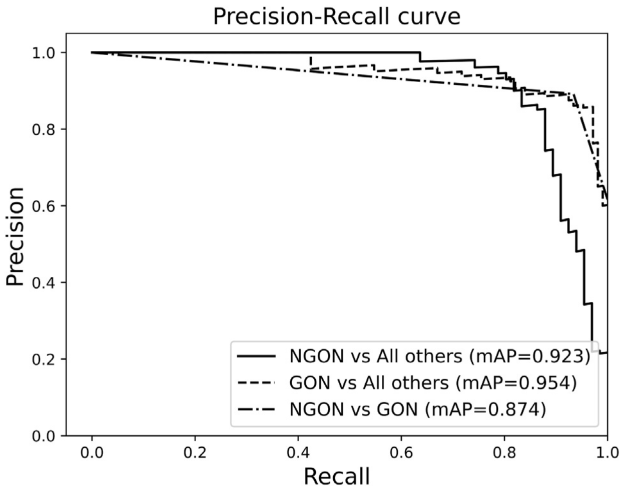

FIGURE 3. Precision-recall curves for the test data set among normal optic discs, nonglaucomatous optic neuropathy with disc pallor (NGON), and glaucomatous optic neuropathy (GON). The area under the P-R curve is presented as the average precision value (mAP).

The diagnostic accuracy of the ResNet-50 model for detecting NGON among normal optic discs and GON images showed a sensitivity of 86.4% and specificity of 99.6%, and the area under the P-R curve showed an average precision of 0.923.

The diagnostic accuracy of the ResNet-50 model to detect GON among normal optic discs and NGON images showed a sensitivity of 92.5% and specificity of 99.5%, with an average precision of 0.954.

The diagnostic accuracy of the ResNet-50 model to specifically differentiate GON from NGON images showed a sensitivity of 93.4% and specificity of 81.8%. The area under the P-R curve for differentiating NGON vs GON showed an average precision of 0.874.

Examples of the CAM visualizing the prediction process of the trained model are shown in Figure 4. As shown in Figure 4, we confirmed that the CAM for each test data was intensively activated on the optic disc, particularly around the cup and neuroretinal rim.

The reasons for false positive findings of GON were extensive areas of peripapillary atrophy and tilted optic discs in 14 cases. False positive findings of NGON were dark background illumination in 3 cases and peripapillary atrophy in 4 cases. Only 1 case with GON showed a false negative result.

DISCUSSION

IN OUR STUDY, THE CNN-BASED DEEP LEARNING MODEL USing ResNet-50 architecture showed excellent performance in classifying NGON and GON as well as normal optic discs in color fundus photographs.

Our study has the following strengths. First, all cases were confirmed and labeled not only by manual interpretation but together with corroborate evidence on visual field testing and OCT. This is important because manual interpretation of the optic disc on fundus photographs are subject to interobserver variability and varying image qualities with different settings.[10,11] Second, we developed a deep learning model to detect not only glaucoma but also NGON, which enhances the utility of the software in mass screening among healthy individuals.[3,4,8–11] Third, we applied batch normalization and dropout methods to improve learning speed and prevent overfitting. As the hidden layer becomes deeper, the possibility of overfitting increases, and the learning time increases. Recently, many methods have been reported to prevent overfitting, and the dropout method is widely used. Dropout is a method that calculates only a part of the weight included in a layer and not the entire weight. Batch normalization is a technique for improving the performance of the artificial neural network through regularization of data distribution in each layer of the neural network by normalizing the activation values.[27] Finally, our deep learning model showed excellent performance in classifying NGON and GON, although we used unstandardized color fundus photographs without controlling image acquisition parameters. This makes our system capable for a broad application on images with variable qualities.

Several reports regarding objective methods for the differentiation of glaucoma and compressive optic neuropathy and/or normal optic discs have been published.[5–7] Hata and associates[5] used quantitative parameters of the optic nerve head measured by OCT and found that cup depths and the distance between the Bruch membrane opening and the anterior surface of the lamina cribrosa were significantly smaller in compressive optic neuropathy than that of glaucoma. However, the cup-to-disc area ratio did not differ between cases of compressive optic neuropathy with a glaucoma-like disc.[5] Nakano and associates[7] estimated the redness of the optic disc neuroretinal rim, the disc color index, using ImageJ software and found that the disc color index of compressive optic neuropathy was significantly smaller than glaucoma particularly in mild stages with an area under the receiver operating characteristic curve above 0.700, which would be useful for differentiation.[7] Ramm and associates[6] quantitatively investigated optic nerve head pallor in patients with primary open-angle glaucoma and healthy subjects using a dual-bandpass transmission filter in the illumination path of the fundus camera of the Dynamic Vessel Analyzer (Imedos Systems UG, Jena, Germany) and recorded 2 monochromatic images at different wavelengths. They found that optic disc pallor values determined by this method were significantly higher in patients with glaucoma compared with healthy subjects. However, these studies used feature extraction and calculation algorithms that were not fully automated.[7]

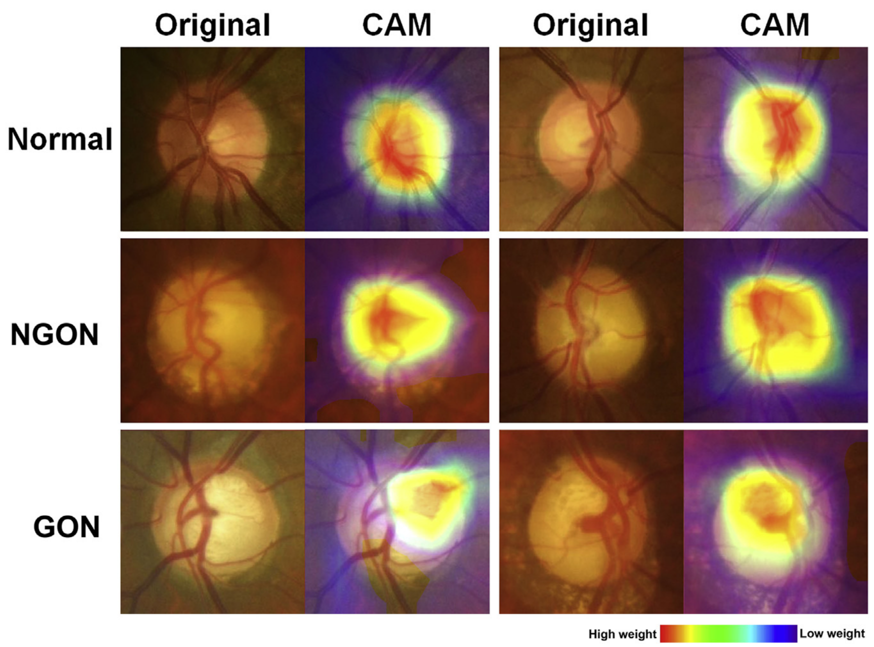

FIGURE 4. Examples of a class activation map (CAM) for prediction of nonglaucomatous optic neuropathy with optic disc pallor (NGON) and glaucomatous optic neuropathy (GON) by the trained model using the test data set.

Until now, a fully automated system for differentiating NGON and GON has not been introduced. In our previous study, we validated an automatic computer-aided diagnosis system for the detection of RNFL defects on fundus photographs showing 86% sensitivity and 75% specificity for detecting RNFL defects in GON and NGON among healthy individuals.[28] We also developed an automated computer-aided diagnosis system for detecting optic disc pallor using color fundus photographs that achieved a sensitivity of 95.3% and a specificity of 96.7%, which was superior to the results of manual detection by individual examiners.[11] These studies were based on explicit features of the optic disc resulting in algorithms that provided excellent detection rates above manual observation.[11,28] However, a combination of structural changes of the optic disc, such as cupping or notching of the rim, RNFL defects, and rim pallor, should be used together to differentiate GON and NGON effectively, which has not been attempted.

Deep learning is a machine learning technique by predictive features taken directly from the images in a large data set of labeled examples. Although deep learning methods have been validated in diabetic retinopathy and age-related macular degeneration,[13–16] this is not easy in optic neuropathy because of the lack of large public data sets and the low reliability of manual detection. Attempts to automatically detect glaucoma from color fundus photographs using deep learning models have shown an overall high sensitivity and specificity.[13,15,17] Li and associates[13] developed a deep learning method for identifying referable GON based on 48 116 fundus photographs showing a robust performance of 95.6% sensitivity, 92.0% specificity, and an area under the curve (AUC) of 0.986. Xiangyu and associates[17] developed a deep learning algorithm for glaucoma detection with combined feature extraction and reported an AUC of 0.831 and 0.887 using the ORIGA and SCES datasets. Liu and associates[15] established a deep learning system for the detection of GON based on 241 032 fundus images comprising definite GON (29 865, 12.4%), probable GON (11 046, 4.6%), and unlikely GON (200 121, 83%). The AUC of the model to detect GON in primary local validation datasets was 0.996 with a sensitivity of 96.2% and specificity of 97.7%.[15] Our results were comparable to these previously reported deep learning systems consisting of a large number of images. Meanwhile, none of these studies compared GON with NGON. The differentiation of NGON and GON based on disc assessment alone may be confusing even for expert ophthalmologists, as in a previous study, the correct diagnosis was identified in 75% of glaucoma and in only 16%-27% of hereditary optic neuropathies.[3] Deep learning models should therefore include both cases of NGON and GON to enhance the utility of the software in mass screening among healthy individuals. Deep learning allows an integral determination of structural changes of the optic disc, and we confirmed that the CAM for each test data was intensively activated on the optic disc, particularly focused on the cup and partly on the neuroretinal rim. This suggests that the characteristic features of cup size and eccentricity, and the color of the neuroretinal rim may be the key to classification, which is consistent with clinical observations.

The most common reasons for false positive findings of GON and NGON in our study were extensive areas of peripapillary atrophy and tilted optic discs. This is similar to the previous deep learning models detecting glaucoma, as approximately half of both false negative and false positive cases were a result of confounding optic disc features caused by high myopia or pathologic myopia.[13,15] These cases require further investigation by eye care professionals and increase the overall workload, but the percentage was minimal in our series and most of them might benefit from further investigation.[13] Physiologic large cupping was also a key feature leading to false positive results in previous models,[13] but this was not found in our study. As only 1 case with GON was classified as normal, the efficacy of our deep learning model as a screening test would be highly useful.

There are certain limitations in our study. First, we included subjects with variable stages of GON and NGON.[5–7] While optic disc pallor is considered the most critical difference between glaucoma and NGON, the discrepancies in structural changes of the optic disc, such as cupping or notching of the rim, RNFL defects, and rim pallor differ according to disease severity.[6] Second, assessment of optic disc morphology may vary according to the brightness and image quality of fundus photographs.[10,29,30] These are the major problems that decrease the reproducibility of image analysis.[10,29,30] However, the best parameters for assessing optic disc color and standardized methods of image acquisition have not been determined.[7,11,30] Moreover, our deep learning model showed excellent performance in classifying NGON and GON without controlling image acquisition parameters. Third, the proportion of each group was not equal in our data set, which included a much higher proportion of GON and NGON compared with the real population. That is why we used the P-R curve instead of the receiver operating characteristic curve, which may be misleading in such circumstances. As this is a major limitation of applying our program for mass screening, further studies using a large dataset are required to verify the efficacy of our program. The number of data in the training set was also insufficient to train the deep learning model and we supplemented the data by augmenting the number of training images.[21,22] Bias from image duplication is possible; however, in order to minimize the impact of imbalanced data between groups during training, we had to afford the bias that would arise from image duplication. Finally, our deep learning model should be validated in different ethnicities because the color of fundus photographs or optic disc morphology may partly depend on the degree of choroidal pigmentation.[11]

In conclusion, we developed an artificial intelligence–based deep learning algorithm for detecting optic disc diseases showing excellent performance in differentiating NGON and GON on color fundus photographs. Our deep learning model may be used in mass screening for the detection of glaucoma and NGON among healthy individuals, as well as assisting the ophthalmologist for differentiating cases that may require urgent neuroimaging and further investigation.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

HEE KYUNG YANG: CONCEPTUALIZATION, METHODOLOGY, Validation, Formal analysis, Writing - original draft. **Young Jae Kim:** Conceptualization, Methodology, Software, Validation, Writing - original draft, Visualization. **Jae Yun Sung:** Data curation. **Dong Hyun Kim:** Data curation. **Kwang Gi Kim:** Conceptualization, Investigation, Resources, Supervision, Project administration, Writing - review & editing, Funding acquisition. **Jeong-Min Hwang:** Conceptualization, Investigation, Resources, Supervision, Project administration, Writing - review & editing, Funding acquisition.

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Drs Y.J. Kim and Yang served jointly as first authors.

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